

Status code and response validation

Status code indicate whether a specific HTTP request has been successful or not. API responds with a status code that tells what happened with the request. There are five types of responses. They are:

- Informational responses (100-199)
- Successful responses (200-299)
- Redirection messages (300-399)
- Client error responses (400-499)
- Server error responses (500-599)

Status Code	What it means	When it Happens
200 OK	Success	GET request successful
201 Created	Resource created	Successful POST request
204 No Content	Success but no body	DELETE or PUT with no return data
400 Bad Request	Invalid request	Wrong JSON format or missing field
401 Unauthorized	No auth or invalid token	Missing or incorrect login credentials
403 Forbidden	Authenticated but not allowed	Token valid but no permission
404 Not Found	Resource not found	Wrong URL or data doesn't exist
500 Internal Server Error	Server issue	Problem on API side

'200 OK' is the checklist for validation. Eg of valid GET response:

```
{
  "userId": 1,
  "id": 1,
  "title": "delectus aut autem",
  "completed": false
}
```

Authentication in APIs

API authentication is the process of verifying the identity of a user who is making an API request, and it is a crucial pillar of API security. There are many types of API authentication, such as HTTP basic authentication, API key authentication, JWT, and OAuth, and each one has its own benefits, trade-offs, and ideal use cases. Nevertheless, all API authentication mechanisms share the goal of protecting sensitive data and ensuring the API is not misused.

Main types of authentication are:

- API Key: Pass a secret key in headers or URL
- Basic Auth: Username + password encoded
- OAuth 2.0: Token from login server

API Test Scenarios

- **GET - Fetch data:**
Check if data can be retrieved successfully
Status code: 200 OK
- **GET - Fetch non-existing data:**
See how the API handles requests for data that doesn't exist
Status code: 404 Not Found
- **POST - Create new resource:**
Verify if new data can be created
Status code: 201 created
- **POST - Missing required fields:**
Ensure validation works correctly
Status code: 400 Bad Request
- **PUT - Update existing resource:**

Update resource with valid data

Status code: 200 OK

- **PUT - Update non-existing resource:**

Try to update a resource that doesn't exist

Status code: 404 Not Found

- **DELETE - Delete existing resource:**

Deletes a valid resource

Status code: 204 No Content

- **DELETE - Delete non-existing resource:**

Try to delete a resource that doesn't exist

Status code: 404 Not Found

